# STOCK PRICE CLASSIFICATION USING MACHINE LEARNING

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## 1. Abstract

This project addresses the limitations of manual stock selection by developing an automated machine learning system designed to identify assets projected to grow by 20 percent or more one year in advance. By integrating 15 years of daily market data from 2010 to 2025 with technical indicators such as MACD and RSI, macroeconomic indicators such as the 10 Year Treasury rate, oil prices and informal indicators such as Big Mac and Buffett Indicator, the study analyzes over 37,000 trading days across ten major United States companies. We evaluated multiple supervised learning models, including Logistic Regression, Random Forest, and XGBoost, utilizing a unified preprocessing pipeline to handle data scaling and imputation. The results indicate that the Logistic Regression model performed best for general portfolio screening, achieving an ROC-AUC of 0.7947 and identifying the 10 Year Treasury rate as the most significant predictor of future stock performance. Additionally, a specialized time series experiment focusing on Apple achieved a remarkable ROC AUC of 0.97 using lagged features, though this approach showed limited generalizability compared to the broader model. Ultimately, the models successfully filter market noise with approximately 80 percent accuracy, providing a scalable and interpretable tool for identifying high conviction investment opportunities.

## 2. Introduction

**Problem Definition and Motivation**

The financial market presents a significant challenge for investors because there are thousands of stocks to choose from, and manual human analysis is often slow, biased, and emotional. Furthermore, asset prices are heavily influenced by invisible macroeconomic factors, such as interest rates and inflation, that are difficult to track manually. The motivation for this project is to address these inefficiencies by developing a machine learning model that identifies "Winning Stocks," defined as assets projected to grow by 20 percent or more one year in advance. Rather than attempting notoriously unreliable exact price predictions, this project aims to filter out market noise and focus on high conviction opportunities.

**Project Inputs and Outputs**

To achieve this, the input to our model consists of a dataset covering 15 years of daily trading history from 2010 to 2025 across ten major United States companies. This input includes internal market data such as adjusted close price, volume, and sector, integrated with external macroeconomic indicators, market context features, seasonality features, and event-related features. (Fig 1.) We then use supervised machine learning models, specifically Logistic Regression, Random Forest, XGBoost and Time-series Logistic Regression to output a binary classification for each asset. The output is a predicted label of either Class 1 (Buy), indicating the stock is expected to outperform the market by at least 20 percent, or Class 0 (Ignore), indicating the stock will fail to meet this growth threshold.

## 3. Related Work

Recent stock-prediction research spans a range of machine-learning approaches, most commonly neural networks, SVMs, and ensemble models. Vijh et al. (2020) compare ANN, Random Forest, and Naïve Bayes models for predicting next-day closing prices using engineered technical features such as moving averages and O/H/L/C-derived indicators. They conclude that model performance varies across stocks and evaluation metrics (RMSE, MAPE), with no single method dominating in all settings. Similarly, Strader et al. (2020) survey machine-learning approaches to financial-market prediction and highlight the prevalence of neural networks and genetic-algorithm hybrids, noting improvements over traditional statistical models but also increased model complexity and reduced interpretability.

Our work differs from these prior studies in three ways: (1) we focus on long-horizon classification predicting whether the 1-year forward return exceeds +20%, rather than short-term price forecasting; (2) we use transparent, interpretable models (logistic regression and Random Forest) rather than relying on black-box neural networks; and (3) our feature set blends technical indicators with broader market-context variables. While prior work demonstrates that complex models can fit short-term patterns, our approach emphasizes interpretability and financial relevance, enabling clearer analysis of feature importance and providing more actionable insights for long-horizon investment decisions.

## 4. Dataset and Features

**Data Sources**

- Historical stock prices from Yahoo Finance
- Macroeconomic indicators from FRED

Our dataset consists of daily historical stock data collected from Yahoo Finance using the yfinance API and macroeconomic indicators from FRED. Each example corresponds to one trading day for one stock. After merging features and constructing labels, we obtained approximately 45147 training samples (before 2023-01-01) and 2500 test samples (after 2023-01-01).

**Data Preprocessing**

We performed several preprocessing steps before training our models. First, we downloaded daily historical data from Yahoo Finance and reshaped into a long format and enriched with sector information. Market indices were handled separately. Next, Macroeconomic Data Integration involved fetching and cleaning various indicators from FRED (10Y Treasury Yield, Oil Price, USD Index), alongside generating synthetic data for the Big Mac Index and Buffett Indicator. All macroeconomic series were resampled to daily frequency and interpolated to fill any gaps. Event & Calendar Features were then created, including binary flags for earnings announcement windows and indicators for the day of the week and month-end. These events and calendar features were consolidated with the macroeconomic data. Following this, we engineered extensive features, calculating a wide array of technical indicators like daily and cumulative returns, SMA divergences, RSI, MACD, and Stochastic Oscillator, along with volume metrics. It also integrated S&P 500 returns and VIX data as market context features. (Appendix 1.) Crucially, a 252-day forward return was calculated to derive the binary Target variable (indicating a >= 20% return). Finally, within the model training phase, a Preprocessing Pipeline handled imputation (median for numerical, most frequent for categorical), scaling (StandardScaler for numerical), and one-hot encoding (for categorical 'Sector' features) before feeding the data to the models.

## 5. Methods

**1. Logistic Regression**

We selected logistic regression as the primary model due to its interpretability and stability.

- Penalty: L1, max_iter: 3000, Class weights: {0:1, 1:3}, Tuned hyperparameter C = 0.5
- ROC-AUC: 0.7947

**2. Random Forest**

- Best hyperparameters: max_depth = 5, min_samples_leaf = 5, n_estimators = 100
- ROC-AUC: 0.7345

**3. XGBoost**

- n_estimators = 100, max_depth = 5 with no hyperparameter tuning
- ROC-AUC: 0.6994

**4. Time-Series Experiment (AAPL Only)**

We conducted an advanced experiment using lagged features for AAPL:

- Lagged 1 to 12 days for each technical feature and target variable
- Logistic regression trained on expanded temporal dataset
- ROC-AUC: 0.97

This confirmed the strong predictive nature of asset-specific time-series features but highlighted potential overfitting when applied to a single stock.

## 6. Experiments and Results Discussion

Because our task is a binary classification problem, we evaluate performance using accuracy, precision, recall, F1-score, and ROC-AUC. ROC-AUC is our primary metric because it measures the model's ability to rank positive events regardless of classification threshold important for financial applications.

**Model Performance Comparison**

| Model | ROC-AUC | Notes |
|---|---|---|
| Logistic Regression | **0.7947** | Best performance and interpretability |
| Random Forest | 0.7345 | Nonlinear interactions but limited depth |
| XGBoost | 0.6994 | Underperformed without tuning |
| Time-Series Experiment | 0.97 | Strong predictive nature of asset-specific time-series features |

Logistic Regression achieved the strongest ROC-AUC, outperforming both tree-based methods. The Random Forest benefited from modeling nonlinear relationships but did not generalize as well as the linear model under our feature set. XGBoost showed the weakest performance, likely due to under-tuned hyperparameters and sensitivity to small signal-to-noise ratios in financial data

**Feature Importance & Insights**

SHAP and model coefficients revealed:

- 10Y Treasury rate is the most influential macroeconomic predictor
- Momentum features (MACD) negatively influence the outcome
- Stocks in technology sector have very strong performance compared to healthcare and energy sectors.
- Buffett indicator negatively influences the outcome, which correlate to its representation of overvalue market when the indicator is high.
- Earning window feature negatively influences the outcome, which suggests investor to sell before releasing. (Fig 1.)

We also ran a time-series experiment using 12-day lagged versions of all technical indicators and the target variable for AAPL. This created a sequence-aware feature set that captures short-term momentum and recent market dynamics. A logistic regression model trained on these lagged features achieved a ROC-AUC of 0.97, substantially higher than the pooled models. This indicates that incorporating recent historical structure can greatly improve prediction performance for a single stock.

## 7. Conclusion

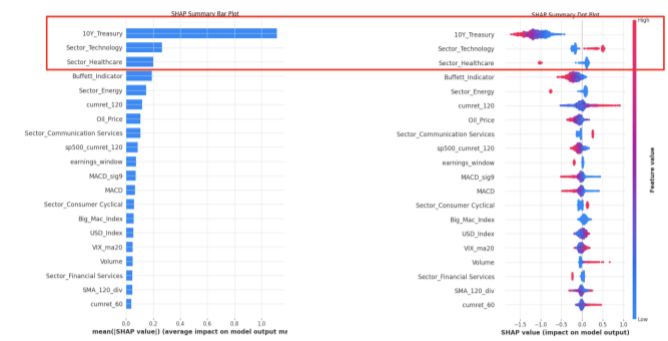
This project demonstrates that stock return classification using machine learning can produce meaningful and interpretable predictions when combining technical, macroeconomic, and engineered features. Logistic Regression emerged as the optimal model, balancing performance and transparency. The 10-year Treasury rate's dominance reinforces how macroeconomic cycles heavily influence long-term equity performance.

The time-series AAPL experiment further proved that lagged covariates significantly enhance predictive power for individual assets. Future work may integrate deep learning, sentiment analysis, and high-frequency signals to improve generalization under volatile regimes.

## 8. Appendices

**Fig 1.**

**SHAP Analysis**




**Model Results:**

| Model | ROC-AUC | Data Scope | Key Strength |
|---|---|---|---|
| Logistic Regression | 0.7947 | All 6 stocks | Best overall + interpretable |
| Random Forest | 0.7345 | All 6 stocks | Strong per-ticker performance |
| XGBoost | 0.6994 | All 6 stocks | Individual stock insights |
| AAPL Time-Series (Manual) | 0.9761 | AAPL only | Exceptional single-stock |
| AAPL Time-Series (Darts) | 0.5278 | AAPL only | Realistic time-series |

**Appendix 1.**

**1. Price-Based Momentum & Trend Features:**

- ret_1d: 1-day percentage return.
- cumret_5: 5-day cumulative return.
- cumret_10: 10-day cumulative return.
- cumret_20: 20-day cumulative return.
- cumret_60: 60-day cumulative return.
- cumret_120: 120-day cumulative return.
- SMA_5_div: Price divergence from 5-day Simple Moving Average.
- SMA_10_div: Price divergence from 10-day Simple Moving Average.
- SMA_20_div: Price divergence from 20-day Simple Moving Average.
- SMA_60_div: Price divergence from 60-day Simple Moving Average.
- SMA_120_div: Price divergence from 120-day Simple Moving Average.
- RSI_14: 14-period Relative Strength Index.
- MACD: Moving Average Convergence Divergence.
- MACD_sig9: 9-period signal line for MACD.
- MACD_hist: MACD Histogram (MACD - MACD_sig9).
- stoch_k14: 14-period Stochastic Oscillator %K.

**2. Volume-Based Features:**

- vol_chg_1d: 1-day percentage change in Volume.
- vol_pct_60: Volume relative to 60-day average volume.

**3. Market Context Features:**

- sp500_ret_1d: 1-day percentage return of S&P 500.
- sp500_cumret_20: 20-day cumulative return of S&P 500.
- sp500_cumret_60: 60-day cumulative return of S&P 500.
- sp500_cumret_120: 120-day cumulative return of S&P 500.
- VIX: CBOE Volatility Index.
- VIX_ma20: 20-day Moving Average of VIX.

**4. Macroeconomic Features:**

- 10Y_Treasury: 10-Year Treasury Constant Maturity Rate.
- Oil_Price: Crude Oil Price (WTI).

- USD_Index: U.S. Dollar Index (Trade Weighted).
- Big_Mac_Index: Synthetic Big Mac Index.
- Buffett_Indicator: Market Capitalization to GDP ratio.

**5. Event & Calendar Features:**

- earnings_window: Binary flag for earnings announcement windows.
- day_of_week: Day of the week (0=Monday, 6=Sunday).
- month_end_flag: Binary flag for month-end period.

**6. Categorical Asset Information:**

- Sector: The industry sector of the asset (e.g., 'Technology', 'Healthcare', 'Financial Services'). This feature is one-hot encoded during preprocessing.

## 9. References


Mehar Vijh, D. C. (2020). Stock Closing Price Prediction using Machine Learning Techniques. *Procedia Computer Science*, pp. 599-606.

Strader, T. J., Rozycki, J. J., ROOT, T. H., & and Huang, Y.-H. J. (2020). Machine Learning Stock Market Prediction Studies: Review and Research Directions. *Journal of International Technology and Information Management:*.